

Computer System

1

RAM:

- Random Access Memory
- Stores running applications
- Temporary
- Volatile

ROM:

- Read Only Memory
- Stores the BIOS and the Boot-up
- Permanent
- Non-volatile

[June 2006]

2

	RAM	ROM
Stores data that the user is currently working on		✓
Storing the system BIOS	✓	
Storing program instructions for computer games	✓	

[June 2007]

3

	Hardware	Software
Word processor		✓
Video camera	✓	
Scanner	✓	
Web page design program		✓

(Nov. 2004)

4

-Windows

-Icon

-Menu

-Pointer

(Specimen 2007)

5

	TRUE	FALSE	
ROM is volatile		✓	[1]
ROM is used to store the startup instructions of a computer	✓		[1]
The data in ROM is easier to change than that in RAM		✓	[1]
ROM is used to store the data the user is currently working on		✓	[1]

(Nov 12 v.2)

6

Three from:

A CLI only allows you to type in commands

With CLI syntax has to be precise

Commands difficult to edit once entered

Have to learn a lot of commands exactly/have to be familiar with the commands [3]

(June13 v.1)

7

Three from

Don't have to type in commands/just use mouse to select options

Easier to change/edit action

Don't have to learn/be familiar with a lot of commands

Easier to open/load programs [3]

(June 13 v.2)

8 Four from:

Read only memory
Can be read from but not written to/can't be changed/edited
Used to store BIOS/start up programs for computer
Used to store software that must not change e.g. games
Non-volatile/permanent (memory)

[4]

(June 13 v.2)

9

	True	False	
Computer programs are examples of software	✓		[1]
A web authoring package is an example of hardware		✓	[1]
An icon is a component of a command line interface		✓	[1]
A mobile phone (cell phone) uses flash memory	✓		[1]

(Nov 12 v.1)

10

Physical components that make up a computer system
Types: Internal and External

[1]

[1]

Three internal devices from:

Motherboard
Random Access Memory (RAM)
Read-Only-Memory (ROM)
Video Card
Sound Card
Internal hard disk drive/solid state drive (HDD/SDD)

[1]

Three external devices from:

Any input device
Any output device
Any external storage devices (e.g. portable hard disk)

[1]

11

Three advantages from:

Very fast to switch on

Fully portable

Touch screen technology means they are simple to use and doesn't need any other input devices

Can use several Apps

Don't generate any heat as they use solid-state technology

Battery life of a tablet is a lot longer

When the power button is pressed it goes into standby but remains connected to the internet, so the user still hears alerts when emails or other notifications are received

[3]

Three disadvantages from:

They often have limited memory or storage compared to a laptop

They can be expensive to run if the internet is being accessed frequently via 3G/4G/5G mobile phone networks

Typing on a touch screen can be slow and error-prone compared to a standard keyboard used by laptop

Transferring of files often has to be done through an 'App Store'

Laptops tend to support more types of file format than tablets, and are also better equipped to run different types of software

[3]

Input Devices

- 1
- | | | |
|---|--------------|-----|
| A | touch screen | [1] |
| B | scanner | [1] |
| C | video camera | [1] |
| D | sensor | [1] |

(Jun 13 V.2)

- 2
- | | |
|---------------------|-----|
| Graphics tablet (1) | [2] |
| Touch screen (1) | |

(June 2009)

- 3
- | | | |
|---------------------|--|-----|
| Mouse | to input voiceovers for a presentation | [1] |
| Joystick | to input a PIN | [1] |
| Optical Mark reader | to select options from a menu | [1] |
| Microphone | to operate a flight simulator | [1] |
| Number pad | to input candidate responses on examination papers | [1] |

(Nov 12 V.2)

- 4
- | | | |
|--|-----------------|---|
| To input text from a printed document | Bar code reader | 1 |
| To input data from a bank cheque | Chip reader | 1 |
| To input data from a bank card | MICR | 1 |
| To input data about a product at a POS | OCR | 1 |

(Jun 11 V.2)

- 5 DVD RAM laser printer magnetic disc
 number pad speakers trackerball [2]

(Jun 13 V.1)

- 6 **Three** from:
Keyboard/typing in data
A bar code (reader)
A magnetic stripe (reader)
Touch screen

Three from:

Scanning bar codes/swiping magnetic stripes/touch screen gives fast data entry/keying in data can be slow

Scanning bar codes/swiping magnetic stripes/touch screen reduces errors/keying in data can lead to data errors

Keyboards/touch screens are robust/bar codes can be flimsy

Magnetic stripes are more robust than bar codes

[6]

(June 2009)

- 7 **Six** from:
Advantages:
No line-of-sight contact is necessary as the tags can be read from a distance of several meters
Robust and reliable technology
Very fast read rate (takes less than 100 milliseconds to respond)
Allows read and write operations to occur (Bidirectional data transfer)
Several RFID tags can be detected at the same time
Disadvantages:
Tag collision may occur
Easy to interrupt or jam as RFID technology uses radio waves
Easy to hack into the data/signal transmitted by the tag
RFID is more expensive than the barcode system [6]

- 8 **Three** from:
Livestock tracking
Retail
Passports
contactless payment
Libraries
Automobiles [3]

9 Four matched pairs from:

Chip reader

Bank/credit card account information/supermarket code

Bar code reader

Information from a product label/product identity number

Electronic scales

Weight of an item

Touch screen

Identification of product

Number pad

Bar code number when bar code reader cannot read bar code/the number of items

Magnetic stripe reader

Information about the customer

[8]

(Jun 13 V.1)

Output Devices

1	Plotter	1
	Robot arm	1

(June 2007)

2		True	False	
	Dot matrix printers produce high quality output		✓	[1]
	Laser printers are very noisy		✓	[1]
	Graph plotters are used when extremely large hard copy is required	✓		[1]
	Inkjet printers are used where continuous stationery is required		✓	[1]

(Nov. 2009)

3	Blu ray disc	Graph plotter	Graphics tablet	1
	Optical mark reader	Projector	Web cam	1

(Jun 10 V.1)

4	to produce very large high quality hard copy	dot matrix printer	1
	to display data from a computer on to a large screen	plotter	1
	to alert mobile phone users to incoming calls	multimedia projector	1
	to output multipart forms	buzzer	1

(Jun 11 V.3)

5	A dot matrix printer	B trackerball	
	C multimedia projector	D inkjet printer	[4]

(Nov 12 V.2)

6 Three from:

Light emitting diodes (LEDs) reach their maximum brightness almost immediately
LEDs give a whiter light, which sharpens the image and makes the colours appear more vivid

LEDs produce a brighter light, which improves the colour definition

LED monitors are much thinner than traditional LCD monitors

LEDs last almost indefinitely which makes the technology more reliable and makes for a more consistent product

LEDs consume very little power, which means they produce less heat as well as using less energy [3]

7 Two from:

Printers that are used to produce solid objects that actually work

Used in computer-aided design (CAD) application

Use additive manufacturing [2]

Four from:

Advantages:

Manufacturing of items has become much easier than ever before

It allows for a rapid prototyping as it can manufacture items relatively quickly

Cost of 3-D printing is less than costs involved in manufacturing a product in the more conventional way

Medical benefits are emerging, such as producing artificial organs, prosthetics and precision-made items for reconstructive surgery

Parts for machinery that are no longer made could now be manufactured using 3-D printers

Disadvantages:

The possibility of producing items that infringe others' copyright

Can be used to produce dangerous items (such as, weapons).

Potential for job losses if this technology takes over from some types of manufacturing [4]

Storage Devices

- 1 memory stick 1
DVD R 1
fixed hard disk drive 1
(June 2007)

- 2 CD ROM drive 1
Memory stick 1
(Nov 2007)

- 3 Fixed hard disc Making backups of file servers
Flash memory card Transferring files from one computer to another
Magnetic tape Storing software sold by a company
Pen drive Storing data which needs to be accessed quickly
DVD ROM Storing photographs in a digital camera [5]
(Nov 11 V.2)

- 4 (a) The most suitable storage medium for making backups of data held on a network server is a magnetic tape [1]
(b) Data about the details of a PIN are stored on a bank card on a chip [1]
(c) An optical disc which can have data updated is a DVD RW [1]
(d) Details of a product are scanned at a POS using a bar code reader [1]

(Nov 11 V.3)

- 5 light pen magnetic disc motor [1]
mouse OCR ROM [1]

(Nov 12 V.2)

6

	True	False	
Computer programs are examples of software	✓		[1]
A web authoring package is an example of hardware		✓	[1]
An icon is a component of a command line interface		✓	[1]
A mobile phone (cell phone) uses flash memory	✓		[1]

(Nov 12 V.1)

7

- (i) A CD ROM is used to store music for sale
- (ii) A blu-ray disc is used to store high definition films
- (iii) A magnetic tape is used to store backups of file servers
- (iv) A flash memory card is used to store photographs in a digital camera

[4]

(Jun 13 V.2)

8

Three from:

Can store more data

Easier to carry/more portable

Majority of computers have USB ports/many school computers don't have CD drives

Speed of access is quicker

Speed of data transfer is quicker

Pen drives are more robust/less prone to damage

[3]

(Nov 11 V.1)

9

Two matched pairs from:

Companies selling their software/games

Cheap method of production/quick to access data

Making personal backups/transferring data (from one computer to another)

Cheap to buy/quicker to retrieve data/expensive to buy tape drives

Downloading/copying films/music

Faster/easier to access individual scenes/tracks/better or higher quality

Two for each matched pair – 4 marks maximum

[4]

(Nov 11 V.3)

10 (a) Three from:

They have no moving parts thus more reliable
 Because there are no moving parts, they are very thin
 They are considerably lighter
 They don't have to get 'up to speed' before they work properly
 They have lower power consumption
 They run much cooler than HDDs
 Data access speed is considerably faster than HDD

[3]

(b) They are considerably lighter

They have lower power consumption than HDDs
 They run much cooler than HDDs

[3]

11 (a) A method of data storage where data is stored on offsite servers, which are owned and managed by a hosting company

[1]

Users can use the cloud to access all their files from any device anywhere in the world

[1]

(b) Data security

Concerned with physical security of the cloud storage

[2]

Data loss

Risk that important and irreplaceable data could be lost from the cloud storage facilities

[2]

(c) Three from:

Customer/client files stored in the cloud can be accessed at any time, from any device, anywhere in the world

There is no need for a customer/client to carry an external storage device with them

The cloud provides the user with remote backup of data with obvious advantages to lower the risk of data loss problem

If a customer/client has a failure of their hard disk or backup device, cloud storage will allow recovery of their data

The cloud system offers almost unlimited storage capacity

[3]

Introduction to databases

1 (a) Numeric (Integer) (1)

(b) Alphanumeric (1)

(c) Boolean (1)

(d) Date (1)

(June 2009)

2	A file is a collection of records/all the students names, addresses, phone numbers and dates of birth	1	
	A field is one item of information/name/address/phone number/date of birth	1	
	A record is all the information about one student/name, address, phone number and date of birth of one student	1	(June 11 V.1)

3	Boolean Meaning Data that is only one of two possible values such as yes/no, true/false Example Gender/Absent or present	1 1	
	Alphanumeric Meaning Any character of the alphabet or number or punctuation mark Example Name/Tutor group/Address/Phone number	1 1	
	Numeric Meaning Must be a digit(s) or decimal number Example Number of siblings/exam marks/number of days present/absent	1 1	(June 11 V.3)

4	4	1	
	7	1	
	The Old Curiosity Shop	1	
	The Moon's a Balloon	1	(Jun 11 V.3)

5	(a)	4	1
	(b)	5	1
	(c)	Relational database	1
	(d)	Two from: Data does not have to be typed in twice/referential integrity. Quicker to enter new data. So fewer errors are likely. So less memory/storage capacity is used. So it is easier to edit data.	2
	(e)	Code	1
	(f)	Book Borrowed	1
	(g)	Cost	1
	(h)	Date	1

(Jun 10 V.1)

6	(a) (i)	Air conditioning (Y/N)	1
	(ii)	One from The field only contains Y or N (two possibilities) It saves time entering data It reduces errors when inputting data It's easier to validate It saves storage space for the file	1
	(b) (i)	Number sold	1
	(ii)	One from: Might need to calculate total of all cars sold Easier to validate Might need to produce a graph Can easily find best selling make	1

(Jun 11 V.1)

7

(a)

Field name	Data type
Model	Text
Camera	Boolean
Free texts	Integer
Monthly payment	Currency

[1]

[1], [1]

[1], [1]

[1]

Three matched pairs from:

Range check on Monthly payment
No less than \$10, no more than \$100

(Invalid) character/type check on 'Free texts' field
Only digits allowed

Presence check on any field
To make sure data has been entered in that field

Consistency check on Monthly payment field and Free texts field
To make sure that number of free texts corresponds to the monthly payment

[6]

(Jun 13 V.2)

8

Three from:

Validation is checking data entered
Is acceptable/reasonable/within a given range
Verification is checking versions of data have been entered identically/is checking data has been copied correctly
From one medium to another (3)

Two from:

Range check
Check digit
Control total
Hash total
Invalid character check/type check
Picture/format check/input mask
Presence check
Existency check
Length check (2)

[5]

(June 2008)

9

Normal data – data within a (given) range/appropriate for that data type
Example – any wage between \$100 and \$500

[1]

[1]

Abnormal data – data outside the range/of the wrong data type
Example – any wage less than \$100 or greater than \$500 or text example

[1]

[1]

Extreme data – data on the boundaries of the range
Example – \$100 or \$500

[1]

[1]

(Jun 13 v.1)

Networks

- 1** Advantages
Two from:
Share hardware
Share software
Share data/information
(Electronic) communication
Users can be more effectively controlled/monitored (2)
- Disadvantages
Two from:
Work can be misused more easily
Viruses more easily spread
Printer queues
Slower access to Internet
If server crashes all computers are affected
Increased security risk (2)
- [4]
(June 2008)
- 2** LAN uses cables to transmit data
WLAN uses wireless technology
- Advantage**
Cheaper as less cabling/new workstations can be sited anywhere/different devices can be connected easily
- Disadvantage**
Slower transfer rates/limited transmission distance/physical obstacles can limit transmission/less security
- 1
1

1

1

(Jun 11 V.1)
- 3** **Two** from:
Switch
Network card
Hub
- One** from
Router
Modem
- 2

1

(Jun 11 V.1)

- 4
- | | |
|---|---|
| A router – so that data can be transferred from one network to another | 1 |
| A browser – to access the (world wide) web /view web pages/surf the net | 1 |
| Email – so that he can send messages to people <u>outside the network/home</u> | 1 |
| An ISP – in order that he can access the internet/email/to provide internet service | 1 |

(Jun 10 V.1)

5

Hub	✓	(1)
Printer		
Scanner		
Switch	✓	(1)
Bridge	✓	(1)
Monitor		

(Nov 11 V.2)

- 6
- Router receives and inspects the data packets sent to it from any computer on any of the networks connected to it [1]
- Router extracts destination's IP address from the data packet [1]
- Router checks the extracted destination's IP address from the data packet against the **routing table** [1]
- The router is then able to route the data packet to reach the appropriate destination router and then to the correct switch/hub [1]
- 7
- Hub broadcasts all the received data packets to every computer on the network [1]
- Switch inspects the data packets and extracts the destination MAC addresses [1]
- Switch then sends the data to the appropriate computer(s) only [1]

8

	Internet	Intranet	
is a network of computer networks	✓		[1]
exists usually within one organisation		✓	[1]
anybody can access it	✓		[1]
can be expanded to become an extranet		✓	[1]

- 9 A **website** is a collection of related webpages [1]
A **web server** is the **computer** that contain web pages of a specific website [1]
A **web browser** is software that allows a user to display a web page on their computer screen [1]

- 10 HTTP:
Two from:
Hypertext transfer protocol
The protocol used when transferring webpages across the internet
Requests and responses are handled by this protocol [2]

HTTPS:
Two from:
Hypertext transfer protocol secure/ Hypertext transfer protocol over SSL
Protocol used when transferring webpages across the internet but with better security than http
Uses SSL to encrypt the requests and responses [2]

FTP:
Two from:
File Transfer Protocol
Used when transferring files from one computer to another computer over the internet [2]

- 11 The internet is the connection of all networks together [1]
WWW consists of massive collection of web pages all connected together using hyperlinks [1]

One from:

World Wide Web is only **part of the Internet**

World Wide Web is a way of making use out of this connection by linking all webpages and websites together [1]

- 12 **Three** from:
Costs are reduced, as there is no need to buy a fax machine, ink or paper
Data transferred is encrypted, which enhances security
Transmissions are sent to an email account, which is password protected
There is no issue of a 'busy signal' prevent the fax being sent [3]

13		True	False	
	A faxed message is more private		✓	1
	An emailed message arrives with the customer more quickly	✓		1
	A faxed document can be used for legal purposes	✓		1
	An email attachment can be edited electronically	✓		1

(Jun 11 V.2)

- 14 (a) **Two** from:
Voice over Internet Protocol (VoIP)
A method that uses the internet to send voice data between the users without using the telephone lines
Converts sound into discrete digital packets that can be sent to their destination via the internet [2]

- (b) **Two advantages** from:
It is either free or at a local rate to anywhere in the world
Much cheaper than video conference if a webcam is used with VoIP
Two disadvantages from:
Quality of the sound when using this technique is completely reliant on the internet connection
Security issues [4]

15 (a) Two from:

Less likely to lose it
May have stronger signal
Display is larger/keyboard is larger
Content is not as limited
Can be faster to access internet
No problem with batteries running out
Has a mouse so is easier to navigate/use

[2]

(b) Two from:

Don't always have access to PC/there may be a power cut
Difficult to carry/not very portable/mobile phone is portable/Cannot access internet except in the house
You have to buy extra hardware/router

[2]

(Nov 13 V.1)

16 Protocol: http
Domain name: google
Domain type: .com
Country code: .eg
Path: googlesearch/pages
Filename: home.htm

[6]

17 Three from:

Causes data to be scrambled/encoded
Requires an encryption key/software to encrypt
Requires a decryption key/encryption software to decrypt
Results in data which is not understandable/readable
Protects sensitive data...
... from being understood if it falls in to the wrong hands

3

(Jun 11 V.3)

18 Strong password:

Password that contains combination of lowercase letters, uppercase letters, numbers and symbols and not a personal info

Example: Aju123ass1RQ@J^@%

Weak password:

Password that doesn't match the strong password criteria

Example: Green

[4]

19 Three matched triples from:

User ID and password

Benefit – each user ID could be unique/only user will know the password/data can only be accessed by person who knows the password/Password can be changed frequently to avoid hackers guessing them/Unsuccessful logins can throw you out of the system

Drawback – user might forget password/keylogging software can be used to intercept it

Biometrics

Benefit – each user has unique biometrics

Drawback – equipment is expensive to buy/may be difficult or expensive to get equipment to user

Magnetic/chip card with PIN

Benefit – Hacker needs to have the card and know the PIN

Drawback – can lose the card/can forget PIN

TAN

Benefit – Always changing so a hacker would not be able to use it even if they intercepted it when user typed it in.

Drawback – need to have card and remember PIN and use it within a short period of time. [9]

(Jun 12 V.1)

20 Phishing

One from:

is using e-mails pretending to be the target's bank

Fake email to get personal details

1

Pharming

One from:

is installing malicious code on a pc or server to redirect to fake websites

redirecting users to websites which look authentic to get personal details

1

Spam (is the sending/receiving of) unsolicited/junk emails

1

(Jun 10 V.1)

21 (a) Two from:

Can lead to denial of services (DoS)/ Uses up valuable bandwidth on the internet, slowing it down

Can be linked to phishing attacks or even spread of computer viruses

Can clog up users' inboxes

[2]

- (b) **Four** from:
- Set the protection level of **junk email filter** to high or to safe lists only
 - Make sure that the junk mail filter is kept up to date
 - Look out for check boxes that are already selected when items are bought online
 - Do not sign up to commercial mailing lists
 - Block images in HTML messages that spammers use as web beacons
 - Do not reply to an email or unsubscribe from a mailing list that you did not explicitly sign up to in the first place
- [4]

- 22 (a) **Three** from:
- It would help to prevent illegal material being posted on the internet
 - It would help to prevent children and other vulnerable groups from being subjected to undesirable websites
 - Harmful material and inappropriate sites can be found easily on the internet especially if using a search engine
 - It would help to stop wrong information being published on websites
- [3]

- (b) **Three** from:
- Material published on websites is already available from other sources
 - It would be expensive to police all websites
 - It would be difficult to enforce rules and regulations on a global scale
 - Policing would be against freedom of information
 - There are laws already exist to deal with those who post illegal material, so no need to police the internet
 - Many things can be offensive to certain people but not to the majority
- [3]

- 23 **Four** from:
- Hackers gaining access to a user's computer through the user of spyware, phishing or pharming
 - The breaking of passwords
 - Visiting a fake website
 - Monitoring internet usage in a public area
 - Hacking cloud storage servers and acquiring sensitive data
- [4]

System Development Life Cycle

1	(a)	Interview shop assistants / customers / Hussain / OR face-to-face question session	[1]
		Can change script to suit circumstances / questions can be explained / repeated / interviewer can interpret body language	[1]
		Observing shop assistants / customers / Hussain / OR watching over the whole process	[1]
		Can gain broader overview of processes / saves workers' time / can gain a more accurate view of what is happening	[1]
		Examining documents used in current system	[1]
		Can follow data flow more easily / informs input / output format design	[1]
		Distributing questionnaires to shop assistants / customers / Hussain / OR using written questions to gather responses	[1]
		Better for busy people / don't have to prearrange appointments / more honest as they are anonymous / takes less time to get all the results (compared with e.g. interview)	[1]
	(b)	Title	[1]
		Author's name	[1]
		Reference / Accession number / ISBN / Bar code number / Publisher / Picture of cover / Dewey number / genre / category / fiction or non-fiction / target age group / picture of author / date published / place published / summary of contents / critique/ no. of copies / price / edition / number of pages / hardback or paperback / language	2 marks for 4 items, 1 mark for 3 items else 0
		Information fills the page AND is clearly a screen form	[1]
		Appropriate sizing for each field	[1]
			[max. 5]
(c)	Three from: Field name Field Type Field Length Key Field Validation Check Index Input mask		[3]

(Nov 2008)

- 2
- (i) Data that is outside the range/not of the right type (1)
Example: a number less than one/a number greater than 12/text (1)
- (ii) Data that is valid/acceptable/within the range/correct type (1)
Example: any number between 1 and 12 inclusive (1)
- (iii) Data that is at the edge of the range (1)
1 or 12 (1)

(Nov 10 V.1)

- 3 Three from:
- The ease of use (of the solution)
- The appropriateness of the solution
- Compare the solution with the original task requirements / old system / design / objectives of system / users needs
- (Identify any) limitations
- (Identify any) improvements which can be made to the system
- Evaluating the users' responses (to the results of testing the system)

[3]

(Nov 2008)

- 4 Direct changeover – new system replaces existing system immediately/overnight (1)
- Parallel running – new system runs alongside/together with existing system (1)
- Phased implementation – new system is implemented part by part (1)
- Pilot running – system is implemented in one branch/one office (at a time) (1)

[4]

(Nov 11 V.1)

5

	Abnormal	Extreme
20		✓
21	✓	
twenty	✓	
0		✓

[4]

(Jun 12 V.1)

6 Technical:

Purpose – (help programmers/analysts) make improvements to system/repair system/maintain system (1)

Two from:

Program listing

Flowchart/algorithm

List of variables

File structure

Purpose of the system

Purpose of the program

Screen layouts

Print formats

Hardware requirements

Software requirements

Sample runs

Known bugs

Error messages

Validation routines (2)

User:

Purpose – help users to learn how to use system (1)

Two from:

How to load software/run software

How to save a file

How to search

How to sort

How to print

How to add records

How to delete/edit records

Troubleshooting guide

Screen layouts (only if not mentioned in technical documentation)

Error messages (only if not mentioned in technical documentation)

Print formats (only if not mentioned in technical documentation)

Hardware requirements (only if not mentioned in technical documentation)

Software requirements (only if not mentioned in technical documentation)

Sample runs (only if not mentioned in technical documentation)

Tutorials (2)

[6]

(June 2008)

- 7 (a)** The age of the target group
The experience of the audience
The expectations of the audience
Knowledge of the audience

[4]

- (b) Interviewing** a cross section of the target group
Giving out **questionnaires** to people in the target group
Carrying out **market research**

[3]

Effects of using ICT

1 Three from:

- Giving software copies to others without the owner's permission
- Making a copy of a disk without the owner's permission
- Selling copies of the software without the owner's permission
- Using the software on a network when the licence does not allow it
- Renting out the software on a network when the licence does not allow it
- Renting out the software/sharing software with others without the permission of the owner
- Using the copyright name on other software
- Amending software/using parts of code in own programs without permission
- Buying pirated software

[3]

(June 2008)

2 (a) RSI (1)

Headaches (1)

[2]

(b) Take regular breaks (1)

Put a screen filter in front of the monitor (1)

[2]

(c) Electrocution (1)

Fire (1)

[2]

(d) Don't overload electrical sockets (1)

Make sure there are no bare wires (1)

[2]

(June 2009)

3

Web logs	✓
Databases	
Models	
Wikis	✓
DTP	
Social networking sites	✓

1

1

1

(Jun 10 V.1)

4

	Blog	Wiki
Usually in some form of chronological order	✓	
Usually many contributors and authors		✓
Usually personal	✓	
Postings tend to be short in length	✓	

[4]

(Jun 12 V.2)

- 5 Problem – electrocution
Prevention – RCB installed/cable insulation
Problem – trailing cables
Prevention – create ducts/cover cables with carpets/description of how length of cabling could be reduced e.g. WiFi
Problem – heavy equipment falling
Prevention – large/sturdy tables
Problem – fire
Prevention – appropriate type of extinguisher/regular maintenance of equipment/ventilation

1 mark for each problem (2 max)

1 mark for each prevention (2 max)

[4]

(June 2008)

- 6 **Two** problems from:
Headaches (from staring at screen continuously)
Eyestrain (from staring at screen continuously)
RSI in fingers (from continual typing/repetitive clicking of mouse)
RSI in wrists (from continual typing/repetitive clicking of mouse)
Backache from sitting in same position all day

Two matching methods from:

Use anti-glare screen (headaches/eyestrain)

Take regular breaks (all)

Use ergonomic keyboard (RSI)

Use wrist rests (RSI)

Use straight backed chair/maintain good posture (backache)

4

(Jun 11 V.1)

- 7 **Four** from:
Microprocessor controlled devices do much of housework
Do not need to do many things manually
Do not need to be in the house when food is cooking
Do not need to be in the house when clothes are being washed
Can leave their home to go shopping/work at any time of the day
Greater social interaction/more family time

(Jun 11 V.2)

- 8** Part time working:
It is working fewer hours per week than full-time staff
- Flexible hours:
It is any working schedule outside the traditional working pattern
OR
The ability to choose the start and finish times, but they must complete a full day's work
- Job sharing:
Full-time job is divided between two part-time workers
- Compressed hours:
Working the full hours for the week for fewer days [4]

- 9** Any data concerning a living person who can be identified by this data [1]
- Three** from:
Name
Address
Date of birth
Medical history
Banking details [1]

- 10 (a) Acts that are set up to protect the rights of the individual about whom the data is obtained, stored and processed [1]
For companies who deal with personal data [1]

- (b) **Four** from:
Data must be fairly and lawfully processed
Data can only be processed for the stated purpose
Data must be sufficient, relevant and not excessive
Data must be accurate
Data must not be kept longer than necessary
Data must be processed according to the data subject's rights
Data must be kept secure
Data must not be transferred to another country unless they also have sufficient protection [4]

- 11 (a) 1 mark for each correct answer
physical safety is the need to prevent accidents with computers
e-safety is the need to take sensible precautions when using the internet [2]

- (b) Any **three** from:
– only use websites recommended by teachers
– only use a student friendly search engine
– only email people already known
– think before opening an email from an unknown person
– never email the school's name or a picture in school uniform
– know how to block and report unwanted users in chat rooms
– never give out any personal information online
– never arrange to meet anyone alone
– always tell an adult first and meet in a public place
– never use real name when playing games online [3]

(Specimen 2016)

- 12 **Four** from:
Requiring to enter a **product key**
The user will be asked to click 'OK'/'I AGREE' to the license agreement before the software continues to install
The original software often comes with a **hologram** sticker informing the purchaser that it is illegal to make copies of this software
Some software will only run if the CD-ROM, DVD-ROM or memory stick is actually in the drive
Putting digital signature on the CD/DVD itself
Some software will only run if a **dongle** is plugged into one of the USB ports [4]

ICT Applications

- 1 Inference engine (1)
Interactive input screen (1)
Knowledge base (1)
Rules base (1)

[4]

(June 2009)

- 2 **Four** from:
Doctors interviewed for requirements
Data is collected from experts
Knowledge base is designed/created
Rule base is designed/created
Inference engine is designed/created
Input Screen is designed/created
Output format is designed/created (4)

Ailments with known diagnoses are used to test the system and results evaluated/does new system match original system specification/interviewing doctors about effectiveness of system (1)

[5]

(June 2008)

3 (a)	Six from: Interactive user screen appears Details of car type are entered Questions about engine problems are asked/on-board computer connected to expert system Answers to questions/engine problems are typed in Inference engine searches Searches the Knowledge base using the Rules (base) Suggested probabilities of faults are output In the form of a report to the mechanic/on screen output	[6]
(b)	Two from: Medical diagnosis Mineral prospecting Chess games Plant identification Animal identification Tax advice Careers advice/guidance Insurance Drug efficacy	[2]

(Nov 2009)

4 (a)	Pressure sensor	✓	[1]
	Proximity sensor		
	Temperature sensor	✓	[1]
	Oxygen sensor		
(b)	Computers work in digital Sensors produce analogue data		[1] [1]
(c)	Five from: Microprocessor continually monitors sensors Data is converted from analogue to digital/ADC is used Compares water level with pre-programmed value If water level reached microprocessor switches off valve Else valve left on/switched on Compares temperature with pre-programmed value If temperature higher microprocessor switches off heater Else heater left on/switched on Compares weight of clothes with pre-programmed value If clothes too heavy microprocessor sounds alarm/stops machine/motor stops running Else cycle continues/motor starts		[5]

(Nov 2009)

- 5 **Three from:**
 If the airplane crashed you wouldn't need to replace it – saving money
 If the airplane crashed nobody is injured
 It is easy to recreate extreme weather conditions
 It is easy to create unusual flying conditions
 Can recreate mechanical/computer failure from previous real situations
 Cheaper fuel costs

3

(Jun 10 V.1)

- 6 **Three from:**
 Reduced cost of wage bill
 Computer readings are more accurate/human errors are reduced
 Readings can be taken more frequently/continuously
 Nurses can get tired and forget to take readings
 Nurses are so busy they might not be able to take readings regularly
 Nurses won't be exposed to contagious diseases
 Automatic warnings can be generated
 Graphs can be produced automatically
 Nurses can be freed up to do other tasks

[3]

(June 2009)

7

	Control	Measurement only
Automatic cookers	✓	
Weather stations		✓
Microwave ovens	✓	
Automatic washing machines	✓	

[4]

(Jun 13 V.2)

8

Computer readings are more accurate.	✓
Students always forget to take readings	
Students might be unavailable to take readings during school holidays.	✓
Computers can record data for later use.	
Computers can analyse the results.	
Computers can react to changes in the conditions immediately.	✓

1

1

1

(Jun 10 V.1)

9

Any **three** from:

- surgical and diagnostic aids
- prosthetics and medical products,
- tissue engineering
- artificial blood vessels
- designs of medical tools and equipment

[3]

(Specimen 2016)

- 10 (a) Two from:**
 Optical Mark Recognition
 Pencil/pen marks are read by scanner/Reader
 Position of marks are identified [2]
 Exam papers/school registers/lottery/multiple choice questionnaires [1]
- (b) Two from:**
 Optical Character Recognition
 Text is read by scanner
 image compared with characters stored in computer
 Converted to text for use with other software [2]
 Utility bill/turnaround documents/word processors/mail/passports/id cards/car number plates [1]
- (c) Two from:**
 Magnetic Ink Character Recognition
 characters read by magnetic reader
 characters compared with characters stored in computer
 Converted to text for entry into system [2]
 Bank cheques [1]

(Nov 11 V.1)

- 11** An algorithm is used to locate and isolate the number plate from the image taken by the camera
 The brightness and contrast of the number plate is first adjusted to ensure that the characters can be clearly read
 Each character on the number plate is then segmented
 Each character is then recognized using OCR software
 The characters are converted into a string of editable text
 The text is then stored on a database [6]
- 12** Cookies are small files or code that are stored on a user's computer [1]
 Sent by a web server to a user's computer and have information about user's preferences [1]
 Web pages visited by the user will be customized according to the user's preferences retrieved from the cookies [1]

- 13 (a) Any three from:**
- Global Positioning System
 - space-based navigation system
 - typically four satellites must be visible to the receiver
 - calculates the distance from a receiver to the satellite
 - calculates the position of the receiver
- [3]

- (b) Any two from:**
- used in cars to calculate routes
 - used by walkers to locate position on hills/mountains
 - used by runners to calculate distance run
 - used by farmers for tractor navigation/soil evaluation/livestock control/yield monitoring
 - used in satellite navigation systems for ships
 - used in tracking aircraft
- [2]

(Specimen 2016)

- 14** To enhance the intelligence and reliability of biometric technology [1]
 Used in fingerprint identification systems to reduce errors
 Checks different fingerprint alignments, which allows the system to match the user's fingerprint in a more intelligent way [2]

OR

Used in Facial-recognition systems to reduce errors due to soft biometric changes
 The system scans a number of faces to learn these soft biometric changes [2]

- 15** The science of using photons (light) and their physical quantum properties to produce a virtually unbreakable encryption system. [1]
 Helps protect the security of data being transmitted over fiber optic cables [1]

- 16** Technology that allows 3-D images/holograms to be produced [1]
Three from:
 Engineering design
 Architecture
 Simulations
 Medical imaging
 Cinema for special effects
 Gaming for special effects
 Advertising
 Holographic televisions
 Holographic computer memories as a new type of optical storage
 Optical computers [3]

Used to develop drones which are unmanned flying devices [1]

Involved in surgical procedures in what is called Robotic Surgery [1]

Two from:

Military use drones in exploration missions

Civilians use drones to survey the landscape in 3-D

Civilians use drones to Investigate weather phenomena

Civilians use drones for search and rescue in natural disasters

Robotic surgery allows surgeons to perform complex procedure with more precision, flexibility and control than the standard surgical techniques [2]

Banking Application

- 1 Producing utility bills (1)
Inputting details of new stock into a library's database (1)
Reading data from bank cheques (1) [3]
(June 2008)

- 2 On-line [1]
Batch [1]
On-line [1]
Batch [1]
(Nov 2008)

- 3 Real Time (1)
Batch (1)
Real Time (1)
Batch (1) [4]
(June 2009)

4	The customer types in the PIN.	1	
	If they are the same the customer is asked which service is required.	3	1
	The customer's account is checked to see if it has sufficient funds.	6/7	1
	The amount is checked against the card limit.	7/6	1
	The PIN number is compared with the PIN stored in the chip.	2	1
	The customer selects required service (cash).	4	1
	If there are sufficient funds and the amount is within the card limit the required notes are issued.	8	1
	The customer is asked how much money they want to withdraw.	5	1

(Jun 10 V.3)

5	The number in stock of the matching record is read	4	1
	Until a match is found with the entered bar code	3	1
	The bar code field in the data file is read record by record	2	1
	The bar code on the product is read by the bar code reader	1	
	The new value of number in stock is written back to the file	8	1
	If it is equal to the re-order number then more goods are automatically re-ordered	7	1
	One is subtracted from the number in stock	5	1
	The number in stock is compared with the re-order number	6	1

(Jun 10 V.1)

6	Four from: Cheques are collected together During the course of the day Cheques are then processed all at once Cheques are processed overnight Bank accounts updated following morning No human intervention	4
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(Jun 11 V.1)

7	Six from: Electronic Funds Transfer at Point of Sale Enables payment for goods at a checkout using credit/debit cards Goods are purchased and bill is calculated Customer inserts card into chip reader Card is checked for validity/reported stolen PIN is entered PIN is compared with that stored on the chip If PIN is OK/verified transaction is authorised If not, customer is asked to re-enter PIN Supermarket computer contacts customer's bank Checks if sufficient funds If sufficient funds, transaction is completed/if not, transaction is rejected Amount deducted from customer's bank account Amount credited to supermarket's bank account The supermarket EFTPOS terminal produces a receipt	[6]
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(Jun 13 V.2)

- 8 The cheque is sent by Smith Bank to a centralized clearing center [1]
The cheque is passed through MICR to automatically read the amount on the cheque and the code line which is consists of account number, sort code and cheque number

[1]

All cheques are sorted using their sort codes

[1]

The data from the cheque, which has been read, is converted into an encrypted file known as **IBDE file** and signed with a digital signature

[1]

All cheques sent by Smith Bank to the centralized clearing center are sent back to Smith Bank after processing them

[1]

Smith Bank then delivers all processed cheques to an **exchange center**

[1]

The **exchange center** passes all cheques to their corresponding paying bank (Hodder Bank in this case)

[1]

Hodder Bank receives the cheque and sends it to its own clearing center to be checked

[1]

At Hodder Bank

- a. The digital signature is first checked
- b. The cheque is passed through Hodder Bank's own MICR to scan the cheque and to make sure that the data matches with that on the IBDE file
- c. The cheque itself is checked to make sure that the details on it is written properly and that the cheque is genuine
- d. Checking if John has enough money in his account to cover the cheque [4]

Based on this information, Hodder Bank finally decides whether to pay John's cheque or to return it unpaid to Smith Bank

[1]